

Analog Integrated Circuit Design

Cadence Tools

Lab 03

Cascode Amplifier

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1 Part 1: Sizing Chart

1.1 Design Specifications

Table 1: Modified design specifications.

Parameter	Value
Technology	0.18 μm CMOS
Supply (V_{DD})	1.8 V
V^* (sweet spot)	200 mV
$I_{D,Q}$ (design current)	68 μA
$g_m r_o$ (intrinsic gain target)	80

Result

68

Lower Limit
Upper Limit

Generate Clear

(a) Random Bias Current=68 μA

Result

80

Lower Limit
Upper Limit

Generate Clear

(b) Random $g_m r_o = 80$

Figure 1: Random parameters.

1.2 Theoretical Background

The intrinsic gain of a MOSFET is given by

$$|A_v| \approx g_m r_o = \frac{2I_D}{V^*} \times \frac{V_A}{I_D} = \frac{2V_A}{V^*} = 80$$

where $V^* = 2I_D/g_m$.

$$V_A = \frac{(g_m r_o) \times V^*}{2} = \frac{80 \times 0.2}{2} = 8 \text{ V}$$

This value of V_A is used later in Part 2 for the hand-analysis output resistance calculations. The required transconductance is calculated directly from V^* :

$$g_m = \frac{2I_D}{V^*} = \frac{2 \times 68 \times 10^{-6} \text{ A}}{200 \times 10^{-3} \text{ V}} = 680 \mu\text{A/V} \quad (1)$$

With the transconductance known, the required output resistance (r_o) and output conductance (g_{ds}) are derived as follows:

$$r_o = \frac{|A_v|}{g_m} = \frac{80}{680 \times 10^{-6} \text{ A/V}} \approx 117.65 \text{ k}\Omega \quad (2)$$

$$g_{ds} = \frac{1}{r_o} \approx 8.5 \mu\text{A/V} \quad (3)$$

1.3 Determining the Channel Length L

Because g_m/g_{ds} (the intrinsic gain) is independent of W at a fixed L and V^* (both g_m and g_{ds} scale proportionally with W), the only way to hit the target $g_m r_o = 80$ is by choosing the right L . A test NMOS ($W = 10\ \mu\text{m}$) was therefore swept in V_{GS} (0 to 0.9 V, 10 mV step, $V_{DS} = V_{DD}/2 = 0.9\text{ V}$) for several trial values of L using a parametric analysis.

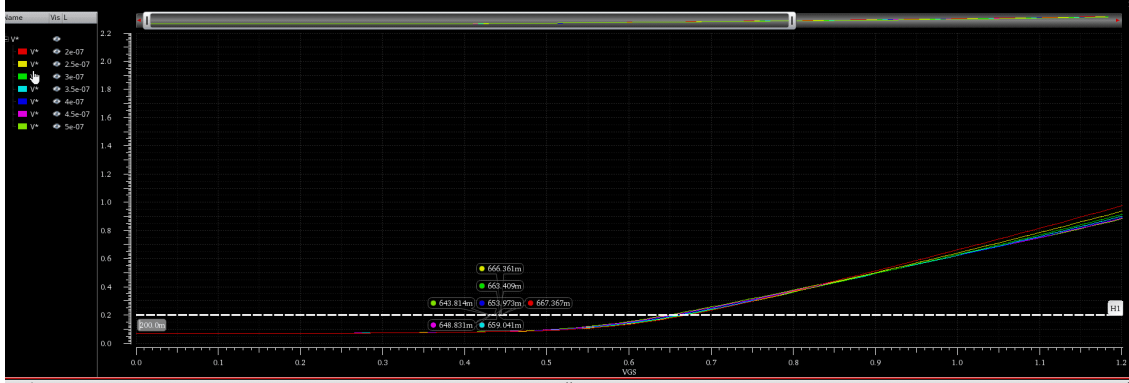


Figure 2: V^* vs. V_{GS} for $L = 0.2\text{--}0.5\ \mu\text{m}$. The 200 mV crossing point shifts with L .

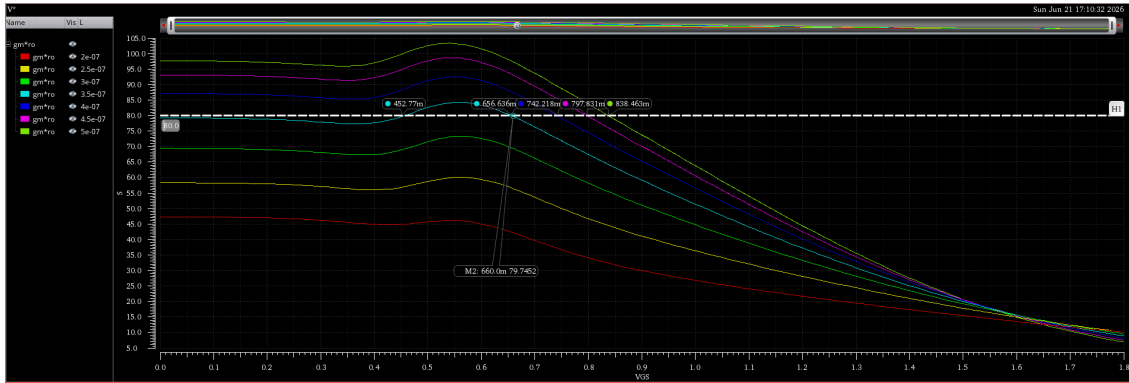


Figure 3: $g_m r_o$ vs. V_{GS} for the same L sweep. The dashed line marks the target gain of 80.

For each L , the V_{GS} at which $V^* = 200\text{ mV}$ was located, and the intrinsic gain $g_m r_o$ was read at that same V_{GS} :

Table 2: Channel length sweep results at $V^* = 200\text{ mV}$.

L	V_{GS} at $V^* = 200\text{ mV}$	$g_m r_o$ at that point
0.20 μm	667 mV	41.805
0.25 μm	666 mV	56
0.30 μm	663 mV	~ 70
0.35 μm	659 mV	79.7
0.40 μm	654 mV	~ 90
0.45 μm	649 mV	~ 97

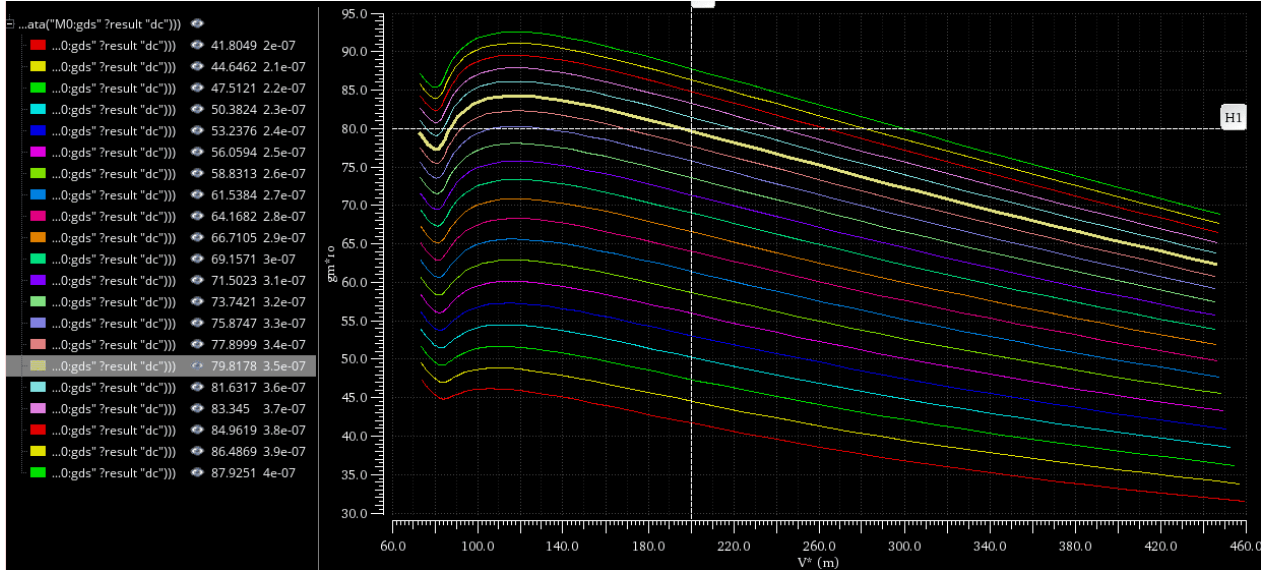


Figure 4: $g_m r_o$ (y axis) vs. V_{star} (x axis) the intersection between the $y=80$ and $x=200\text{m}$ at $L=350\text{nm}$

$L = 0.35\text{ }\mu\text{m}$ gives the closest match to the target intrinsic gain (79.7 vs. 80, within 0.4%), so it is selected as the design channel length:

$$L = 0.35\text{ }\mu\text{m}, \quad V_{GSQ} \approx 659\text{ mV}$$

1.4 Sizing W for $I_D = 68\text{ }\mu\text{A}$

With L fixed at $0.35\text{ }\mu\text{m}$, the chart values at $W = 10\text{ }\mu\text{m}$, $V_{GS} = 659\text{ mV}$ are:

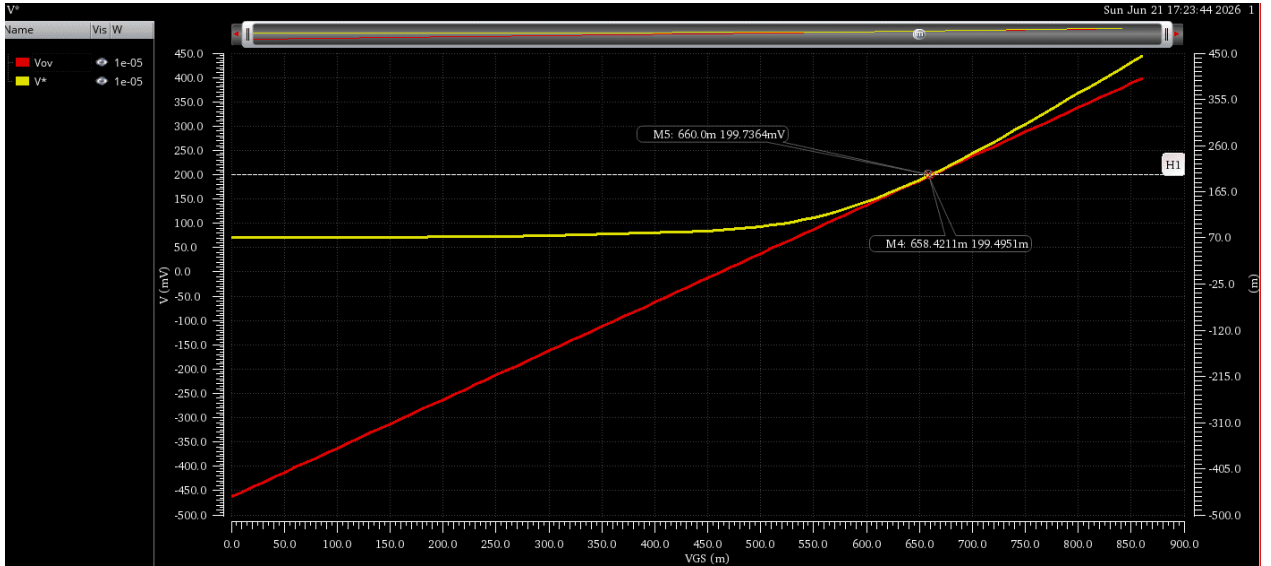


Figure 5: $V^* = 2I_D/g_m$ (Yellow) and $V_{ov} = V_{GS} - V_{TH}$ (Red) vs. V_{GS} for the NMOS with $W = 10\text{ }\mu\text{m}$, $L = 0.35\text{ }\mu\text{m}$.

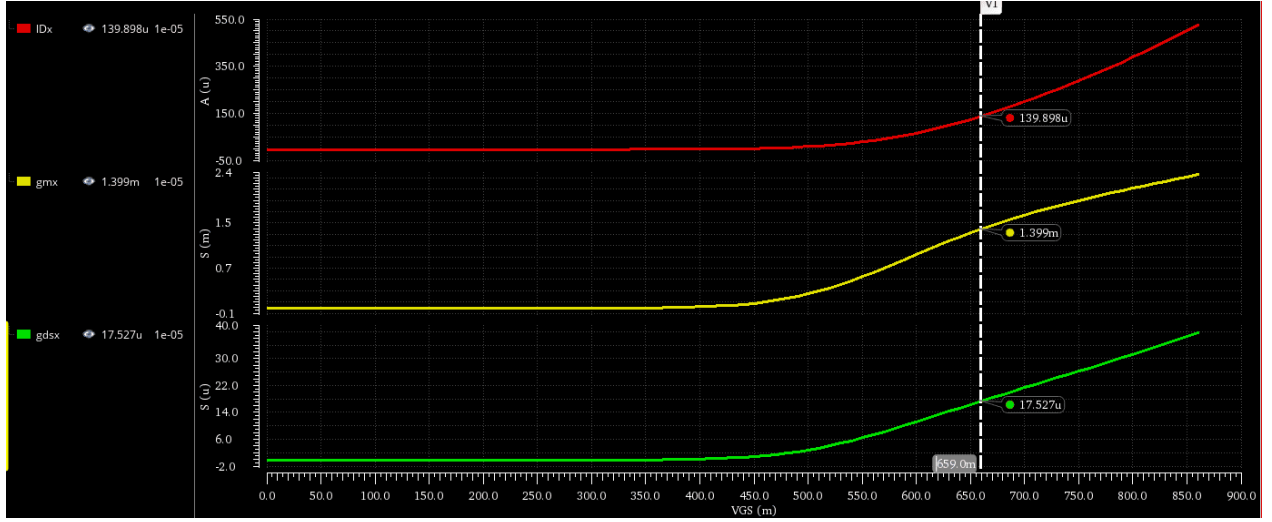


Figure 6: I_D (Green) , g_m (Yellow) , and g_{ds} (Red) vs. V_{GS} for the reference NMOS ($W = 10 \mu m$, $L = 0.35 \mu m$, $V_{DS} = V_{DD}/2$).

Table 3: Values read from the sizing chart at $V_Q^* = 0.2 V$, $W_{ref} = 10 \mu m$, $L = 0.35 \mu m$.

Quantity	Symbol	Value
Reference drain current	I_{DX}	139.9 μA
Reference transconductance	g_{mX}	1.399 mS
Reference output conductance	$g_{ds,X}$	17.527 μS
$V_{ov,Q}$	$V_{ov,Q}$	199.7 mV
Corresponding V_{GS}	V_{GSQ}	659 mV

Since I_D , g_m , and g_{ds} are all proportional to W at fixed V^* and L , ratio and proportion gives:

$$W = 10 \mu m \times \frac{I_{D,Q}}{I_{DX}} = 10 \mu m \times \frac{68 \mu A}{139.90 \mu A} = 4.86 \mu m$$

$$g_{mQ} = g_{mX} \times \frac{I_{D,Q}}{I_{DX}} = 1.399 mA/V \times 0.4861 = 680 \mu A/V$$

$$g_{dsQ} = g_{dsX} \times \frac{I_{D,Q}}{I_{DX}} = 17.527 \mu A/V \times 0.4861 = 8.52 \mu A/V \Rightarrow r_{oQ} = \frac{1}{g_{ds,Q}} \approx 117.37 K\Omega$$

As a consistency check, $V_A = I_{DX}/g_{dsX} = 7.98 V$, matching the 8 V predicted from the $g_m r_o = 80$ spec, confirming the design is self-consistent.

1.4.1 Gain Verification

The actual gain including r_o :

$$A_v = -g_m r_o = -680 \mu \times 117.4 k\Omega = -79.832 \approx -80 \quad (4)$$

The parameters are correct!

Table 4: Final Part 1 design results, used for M1, M2, M3 in Part 2. Final design parameters from Part 1 sizing chart ($0.18\text{ }\mu\text{m}$ CMOS, $L = 0.35\text{ }\mu\text{m}$, $I_{DQ} = 68\text{ }\mu\text{A}$, $|A_v| = 80$).

Parameter	Symbol	Value
Transistor Width	W	$4.86\text{ }\mu\text{m}$
Channel Length	L	$0.35\text{ }\mu\text{m}$
Transconductance	g_m	$680\text{ }\mu\text{S}$
Output Conductance	g_{ds}	$8.52\text{ }\mu\text{S}$
Output Resistance	r_o	$117.4\text{ k}\Omega$
Gate-Source Bias Voltage	V_{GSQ}	659 mV

2 Part 2: Cascode for Gain

2.1 Circuit Construction

The sizing from Part 1 $W = 4.86\text{ }\mu\text{m}$, $L = 0.35\text{ }\mu\text{m}$, $I_B = 68\text{ }\mu\text{A}$, $C_L = 1\text{ pF}$, $V_{DD} = 1.8\text{ V}$.

A feedback resistor R_f connects each output back to its input transistor's gate ($R_{f,DC} = 1\text{ m}\Omega$, $R_{f,AC} = 1\text{ T}\Omega$ — a short at DC, an open at AC), while a source resistor R_s connects the signal source to the same gate with the opposite values ($R_{s,DC} = 1\text{ T}\Omega$, $R_{s,AC} = 1\text{ m}\Omega$). At DC this makes M1 and M2 diode-connected (forcing saturation); at AC the feedback path disappears and the signal source drives the gate directly. M3's gate is tied to a plain DC source V_B .

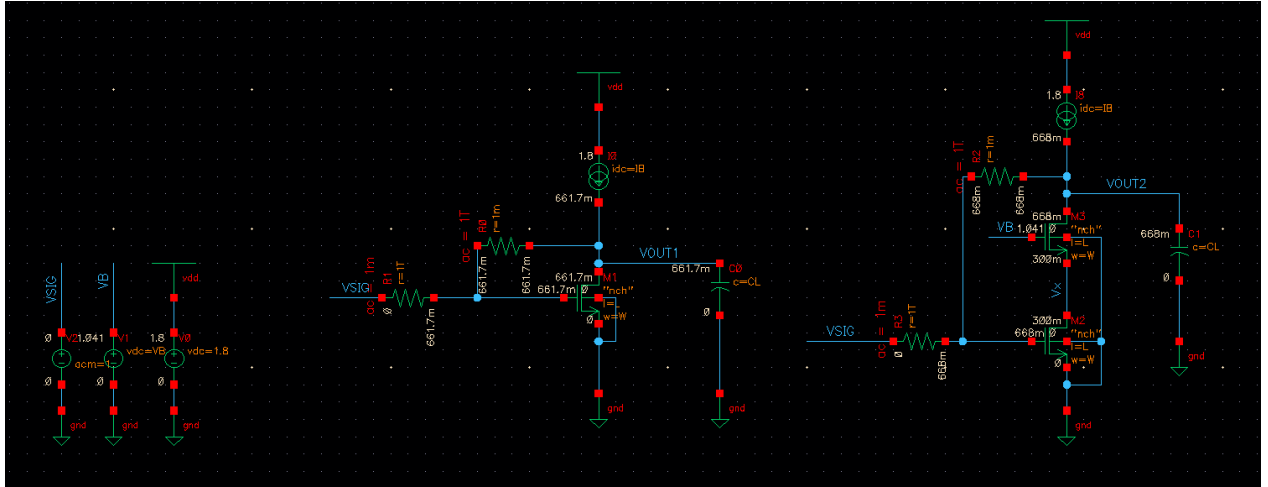


Figure 7: CS amplifier (M1, left) and cascode amplifier (M2, M3, right) schematics, with the DC/AC dual-value feedback resistors (R_1/R_0 and R_3/R_2) and node labels VOUT1, VOUT2, VX annotated.

2.2 Determining V_B

V_B was chosen such that M2 (the cascode amplifier's input device) has $V_{DS} \approx V^* + 100\text{ mV} = 300\text{ mV}$, leaving the input device just enough headroom to stay in saturation while maximizing the swing available to M3. A DC sweep of V_B from 0 to 1.8 V was performed while plotting V_{DS} of M2:

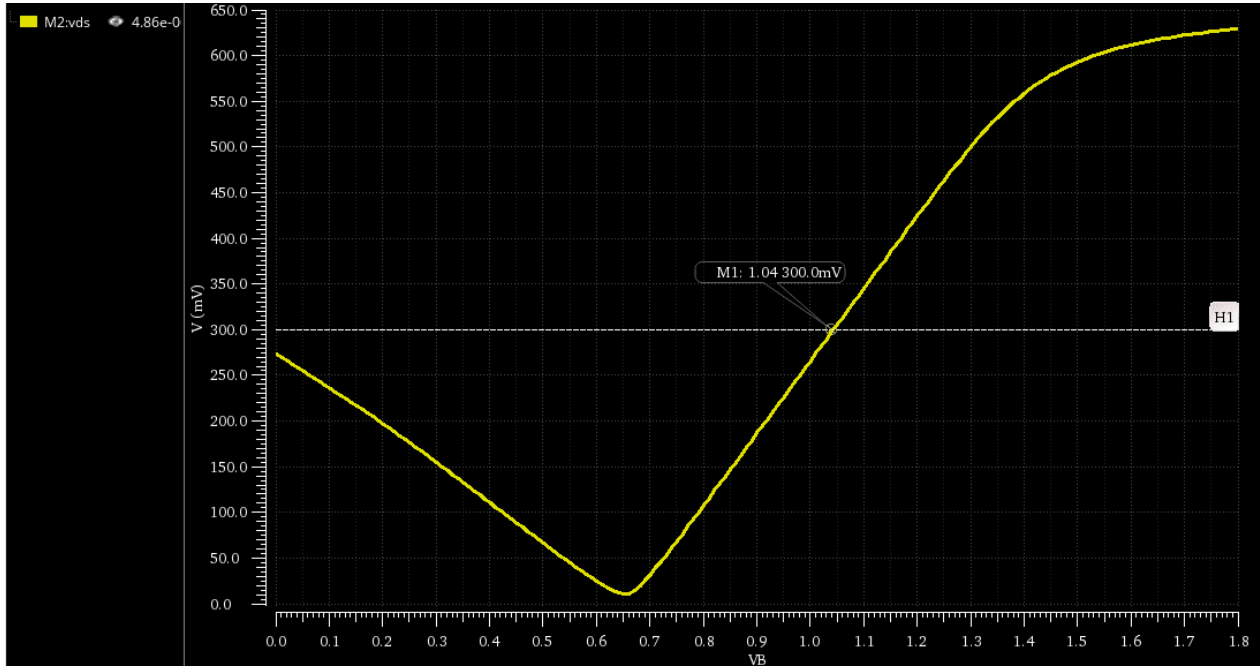


Figure 8: V_{DS} of M2 vs. V_B . The 300 mV crossing point sets the cascode bias voltage.

The curve crosses 300 mV at $V_B = 1.041$ V (simulated $V_{DS}(M2) = 300$ mV), which was used as the fixed cascode bias for the remainder of Part 2.

2.3 DC Operating Point Results

The DC OP simulation was run with $V_B = 1.041$ V, $I_B = 68$ μ A, $W = 4.86$ μ m, $L = 0.35$ μ m for all three transistors.



Figure 9: Variables used in DC analysis

ITi:LAB03_2:1	OP("/M1" "id")	68u	ITi:LAB03_2:1	OP("/M2" "id")	68u	ITi:LAB03_2:1	OP("/M3" "id")	68u
ITi:LAB03_2:1	OP("/M1" "vgs")	661.7m	ITi:LAB03_2:1	OP("/M2" "vgs")	668m	ITi:LAB03_2:1	OP("/M3" "vgs")	741m
ITi:LAB03_2:1	OP("/M1" "vds")	661.7m	ITi:LAB03_2:1	OP("/M2" "vds")	300m	ITi:LAB03_2:1	OP("/M3" "vds")	368m
ITi:LAB03_2:1	OP("/M1" "vth")	461.8m	ITi:LAB03_2:1	OP("/M2" "vth")	464.7m	ITi:LAB03_2:1	OP("/M3" "vth")	547.8m
ITi:LAB03_2:1	OP("/M1" "vdsat")	153.3m	ITi:LAB03_2:1	OP("/M2" "vdsat")	155.6m	ITi:LAB03_2:1	OP("/M3" "vdsat")	158.7m
ITi:LAB03_2:1	OP("/M1" "gm")	678.1u	ITi:LAB03_2:1	OP("/M2" "gm")	662.9u	ITi:LAB03_2:1	OP("/M3" "gm")	675.8u
ITi:LAB03_2:1	OP("/M1" "gds")	9.337u	ITi:LAB03_2:1	OP("/M2" "gds")	18.57u	ITi:LAB03_2:1	OP("/M3" "gds")	14.33u
ITi:LAB03_2:1	OP("/M1" "gmb")	177.5u	ITi:LAB03_2:1	OP("/M2" "gmb")	173.8u	ITi:LAB03_2:1	OP("/M3" "gmb")	159.8u
ITi:LAB03_2:1	OP("/M1" "cdb")	-3.705f	ITi:LAB03_2:1	OP("/M2" "cdb")	-4.063f	ITi:LAB03_2:1	OP("/M3" "cdb")	-3.716f
ITi:LAB03_2:1	OP("/M1" "cgd")	-2.323f	ITi:LAB03_2:1	OP("/M2" "cgd")	-2.363f	ITi:LAB03_2:1	OP("/M3" "cgd")	-2.34f
ITi:LAB03_2:1	OP("/M1" "cgs")	-12.68f	ITi:LAB03_2:1	OP("/M2" "cgs")	-12.7f	ITi:LAB03_2:1	OP("/M3" "cgs")	-12.62f
ITi:LAB03_2:1	OP("/M1" "csb")	-7.012f	ITi:LAB03_2:1	OP("/M2" "csb")	-7f	ITi:LAB03_2:1	OP("/M3" "csb")	-6.258f
ITi:LAB03_2:1	OP("/M1" "region")	2	ITi:LAB03_2:1	OP("/M2" "region")	2	ITi:LAB03_2:1	OP("/M3" "region")	2

(a) M1 Parameters

(b) M2 Parameters

(c) M3 Parameters

Figure 10: Operating point parameters for transistors M1, M2, and M3.

Table 5: DC operating-point parameters for M1 (CS), M2 (cascode input), M3 (cascode device).

Parameter	M1	M2	M3
I_D (μA)	68	68	68
V_{GS} (mV)	661.7	668.0	741.0
V_{DS} (mV)	661.7	300.0	368.0
V_{TH} (mV)	461.8	464.7	547.8
V_{DSAT} (mV)	153.3	155.6	158.7
g_m ($\mu\text{A}/\text{V}$)	678.1	662.9	675.8
g_{ds} ($\mu\text{A}/\text{V}$)	9.337	18.57	14.33
g_{mb} ($\mu\text{A}/\text{V}$)	177.5	173.8	159.8
C_{db} (fF)	3.705	4.063	3.716
C_{gd} (fF)	2.323	2.363	2.340
C_{gs} (fF)	12.68	12.70	12.62
C_{sb} (fF)	7.012	7.000	6.258
Region	Sat (2)	Sat (2)	Sat (2)

Table 6: DC node voltages and bias.

Node	Value
V_{OUT1} (M1 drain)	661.7 mV
V_{OUT2} (M3 drain)	668.0 mV
V_X (M2 drain / M3 source)	300.0 mV
V_B (M3 gate)	1.041 V

2.3.1 Design Validation

As a check on the Part 1 design methodology, $V_{ov} = V_{GS} - V_{TH}$ was computed for each transistor and compared to the target $V^* = 200\text{ mV}$:

Table 7: Effective overdrive at the operating point, compared to the 200 mV design target.

	M1	M2	M3
$V_{ov} = V_{GS} - V_{TH}$ (mV)	199.9	203.3	193.2

All three sit within 7 mV of the 200 mV target despite operating at different V_{DS} and (for M3) a different V_{SB} than the Part 1 calibration testbench ($V_{DS} = V_{DD}/2$, $V_{SB} = 0$) — confirming the V^* -based chart sizing transfers well to the real circuit.

2.3.2 Saturation Check

All three transistors report `region = 2` (saturation), and this is confirmed by comparing V_{DS} to V_{DSAT} :

Table 8: Saturation margin ($V_{DS} - V_{DSAT}$).

	M1	M2	M3
V_{DS} (mV)	661.7	300.0	368.0
V_{DSAT} (mV)	153.3	155.6	158.7
Margin (mV)	508.4	144.4	209.3

All margins are positive, confirming saturation.

Comments

Q: Check that all transistors operate in saturation.

Confirmed above: all three report `region=2`, and $V_{DS} > V_{DSAT}$ for each (Table 5).

Q: Do all transistors have the same V_{TH} ? Why?

No. M1 ($V_{TH} = 461.8$ mV) and M2 ($V_{TH} = 464.7$ mV) are nearly identical, since both have their source grounded ($V_{SB} = 0$) — the small 2.9 mV difference is a second-order effect from their different V_{DS} . M3, however, has a noticeably higher $V_{TH} = 547.8$ mV because its source is *not* grounded — it sits at $V_X = 300.0$ mV due to the cascode connection, giving $V_{SB}(M3) = 300.0$ mV. This non-zero V_{SB} raises V_{TH} through the body effect, $V_{TH} = V_{TH0} + \gamma(\sqrt{V_{SB} + \phi} - \sqrt{\phi})$. This is exactly why M3 requires a much higher V_{GS} (741.0 mV) than M1/M2 to deliver the same 68 μ A at a similar overdrive.

Q: Relation between g_m and g_{ds} ?

$g_m \gg g_{ds}$ for all three transistors (ratios of ≈ 72.6 , 35.7, and 47.2 for M1, M2, M3 respectively). This large disparity is precisely what allows the transistor to provide voltage gain ($A_v \approx -g_m/g_{ds}$ for a current-source-loaded stage).

Q: Relation between g_m and g_{mb} ?

$g_m > g_{mb}$ for all three ($g_{mb}/g_m \approx 0.262, 0.262, 0.237$ for M1, M2, M3).

Q: Relation between C_{gs} and C_{gd} ?

$C_{gs} \gg C_{gd}$ in magnitude for all three ($\approx 5.4\times$ larger). In saturation, $C_{gs} \approx \frac{2}{3}C_{ox}WL$ is a real intrinsic channel capacitance, while C_{gd} in saturation is dominated only by the small gate-drain overlap capacitance.

Q: Relation between C_{sb} and C_{db} ?

$C_{sb} > C_{db}$ in magnitude for all three ($\approx 1.7\text{--}1.9\times$ larger). Junction capacitance decreases with increasing reverse bias across the junction. Since the drain sits at a higher DC voltage than the source (for the same grounded bulk), the drain-bulk junction is more reverse-biased than the source-bulk junction, giving it a smaller depletion capacitance — hence $C_{db} < C_{sb}$.

2.4 AC Analysis

2.4.1 Hand Analysis

CS Amplifier (M1). The current-source load is ideal, so the only resistance at VOUT1 is M1's own output resistance:

$$r_{o1} = \frac{1}{g_{ds1}} = \frac{1}{9.337 \mu\text{A/V}} = 107.10 \text{ k}\Omega \quad (5)$$

$$A_{v1} = -g_{m1} r_{o1} = -678.1 \mu\text{A/V} \times 107.10 \text{ k}\Omega = -72.63 \text{ (37.22 dB)} \quad (6)$$

The dominant pole is set by C_L at the output (the parasitic C_{db1}, C_{gd1} are only a few fF, negligible next to $C_L = 1$ pF):

$$BW_1 = \frac{1}{2\pi r_{o1}(C_L + C_{gd1} + C_{db1})} = \frac{1}{2\pi \times 107.10 \text{ k}\Omega \times (1 \text{ pF} + 2.323 \text{ fF} + 3.705 \text{ fF})} = 1.4775 \text{ MHz} \quad (7)$$

$$GBW_1 = |A_{v1}| \cdot BW_1 = \frac{g_{m1}}{2\pi C_L} = \frac{678.1 \mu\text{A/V}}{2\pi \times 1 \text{ pF}} = 107.95 \text{ MHz} \quad (8)$$

Since $|A_{v1}| \gg 1$, $UGF_1 \approx GBW_1 = 107.95 \text{ MHz}$.

Cascode Amplifier (M2, M3). The output resistance seen at VOUT2 is the standard cascode boosting formula, with r_{o2} acting as source degeneration for M3:

$$R_{out} = r_{o2} + r_{o3} + (g_{m3} + g_{mb3}) r_{o2} r_{o3} \quad (9)$$

with $r_{o2} = 1/g_{ds2} = 53.85 \text{ k}\Omega$, $r_{o3} = 1/g_{ds3} = 69.78 \text{ k}\Omega$, $g_{m3} + g_{mb3} = 835.6 \mu\text{A/V}$:

$$R_{out} = 53.85k + 69.78k + (835.6\mu)(53.85k)(69.78k) = 3.264 \text{ M}\Omega \quad (10)$$

$$A_{v2} = -g_{m2} R_{out} = -662.9 \mu\text{A/V} \times 3.264 \text{ M}\Omega = -2164.7 \text{ (66.71 dB)} \quad (11)$$

$$BW_2 = \frac{1}{2\pi R_{out}(C_L + C_{gd3} + C_{db3})} = \frac{1}{2\pi \times 3.264 \text{ M}\Omega \times (1 \text{ pF} + 2.340 \text{ fF} + 3.716 \text{ fF})} = 48.46 \text{ kHz} \quad (12)$$

$$GBW_2 = \frac{g_{m2}}{2\pi C_L} = 105.50 \text{ MHz}, \quad UGF_2 \approx GBW_2 \quad (13)$$

Note that GBW depends only on the *input* transistor's g_m and on C_L – it is independent of R_{out} , which is why $GBW_1 \approx GBW_2$ despite the output resistances differing by $\sim 30\times$.

2.4.2 Simulation Results

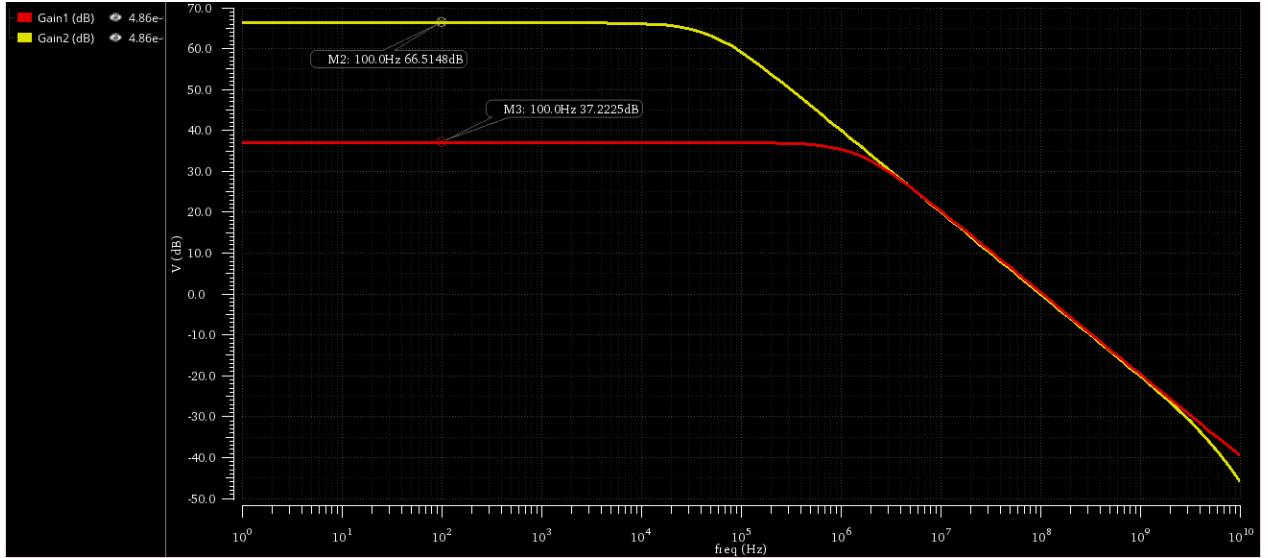


Figure 11: Magnitude in dB (Gain 1, red) and cascode amplifier (Gain 2, yellow).

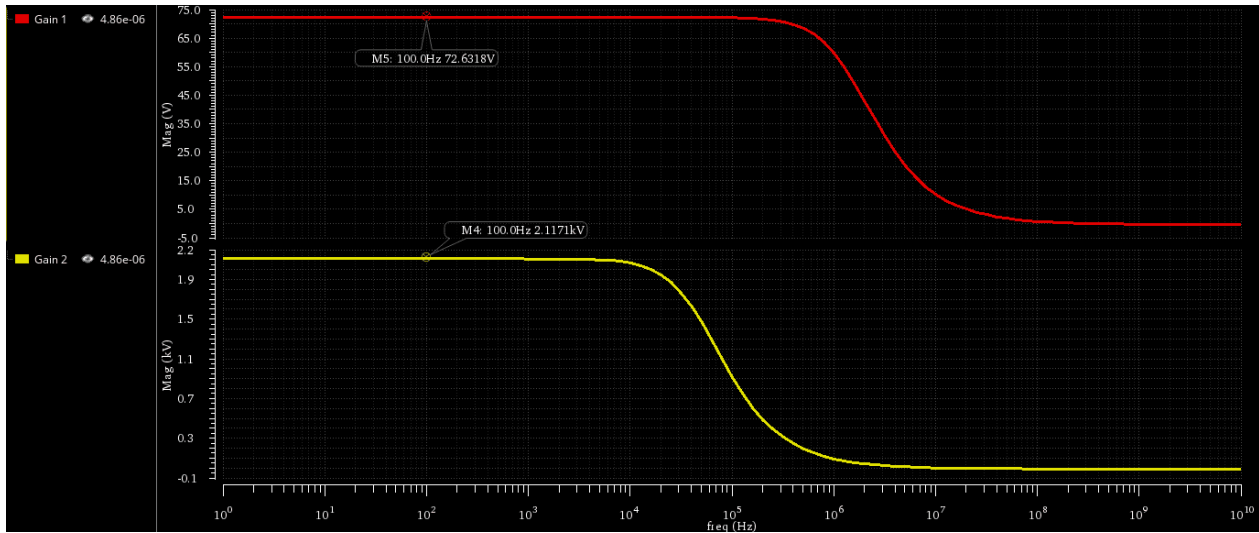


Figure 12: Linear magnitude (V/V) (Gain 1, red) and cascode amplifier (Gain 2, yellow).

2.4.3 Comparison: Simulation vs. Hand Analysis

Test	Output	Nominal
ITi:LAB03_2:1	Gain 1	
ITi:LAB03_2:1	Gain1 (dB)	
ITi:LAB03_2:1	DC Gain1 (dB)	37.22
ITi:LAB03_2:1	DC Gain1	72.63
ITi:LAB03_2:1	BW 1	1.477M
ITi:LAB03_2:1	GBW 1	107.5M
ITi:LAB03_2:1	UGF 1	108.5M
ITi:LAB03_2:1	Gain 2	
ITi:LAB03_2:1	Gain2 (dB)	
ITi:LAB03_2:1	DC Gain2 (dB)	66.51
ITi:LAB03_2:1	DC Gain2	2.117k
ITi:LAB03_2:1	BW 2	48.41k
ITi:LAB03_2:1	GBW 2	102.7M
ITi:LAB03_2:1	UGF 2	103.2M

Figure 13: Circuit parameters values

Table 9: CS amplifier (M1): simulation vs. hand analysis.

Quantity	Hand Analysis	Simulation	Error (%)
DC Gain (V/V)	72.63	72.63	0.00
DC Gain (dB)	37.22	37.22	0.00
BW	1.4775 MHz	1.477 MHz	0.03
GBW	107.95 MHz	107.5 MHz	0.42
UGF	107.95 MHz	108.5 MHz	0.51

Table 10: Cascode amplifier (M2, M3): simulation vs. hand analysis.

Quantity	Hand Analysis	Simulation	Error (%)
DC Gain (V/V)	2164.7	2117	2.25
DC Gain (dB)	66.71	66.51	0.30
BW	48.46 kHz	48.41 kHz	0.10
GBW	105.50 MHz	102.7 MHz	2.73
UGF	105.50 MHz	103.2 MHz	2.23

Comments

Q: Comment on the results.

The simulation results closely match the theoretical hand calculations, especially for the CS stage (error <0.6% on all metrics). The cascode configuration successfully boosts the DC gain by significantly increasing the output impedance (roughly by a factor of $g_{m3}r_{o3}$ on top of r_{o2}). Consequently, because the dominant pole is located at $\approx 1/(R_{out}C_L)$, the bandwidth of the cascode amplifier is reduced by approximately that same factor:

$$\frac{A_{v2}}{A_{v1}} = \frac{2117}{72.63} \approx 29.1 \times \text{gain increase}, \quad \frac{BW_1}{BW_2} = \frac{1.477M}{48.41k} \approx 30.5 \times \text{bandwidth decrease}$$

Because the gain increases and the bandwidth decreases by almost the same proportion, the Gain-Bandwidth Product (GBW) remains relatively constant between both topologies ($\approx 105\text{--}108\text{ MHz}$), demonstrating the fundamental trade-off between gain and bandwidth in single-stage amplifiers. The slightly larger discrepancy on the cascode's GBW/UGF ($\sim 2.2\text{--}2.7\%$, vs. $<0.6\%$ for the CS stage) is expected, since the cascode's hand formula compounds three measured small-signal parameters (g_{m3} , g_{mb3} , and both r_o 's) into a single product, amplifying any individual parameter-extraction error, whereas the CS stage's gain depends on only two parameters.

3 Part 3 [Optional]: Cascode for BW

3.1 Circuit Construction

The current-source loads of Part 2 were replaced with load resistors R_D , eliminating the need for the DC/AC feedback-resistor biasing trick: the resistor itself sets the DC operating point, so the input transistors' gates are driven directly from a DC-biased signal source ($V_{GS} = 659\text{ mV}$, the Part 1 design point) through a $10\text{ M}\Omega$ source resistance, with no feedback path required. C_L was reduced from 1 pF to 1 fF to make the bandwidth set primarily by $R_{out} \parallel R_D$ and the device parasitics rather than by a large external load cap.

Sizing R_D . With the same bias current as before ($I_{D,Q} \approx 68\text{ }\mu\text{A}$, since V_{GS} is unchanged) and targeting a $V_{DD}/2$ drop across the resistor:

$$R_D = \frac{V_{DD}/2}{I_{D,Q}} = \frac{0.9\text{ V}}{68\text{ }\mu\text{A}} = 13.235\text{ k}\Omega \quad (14)$$

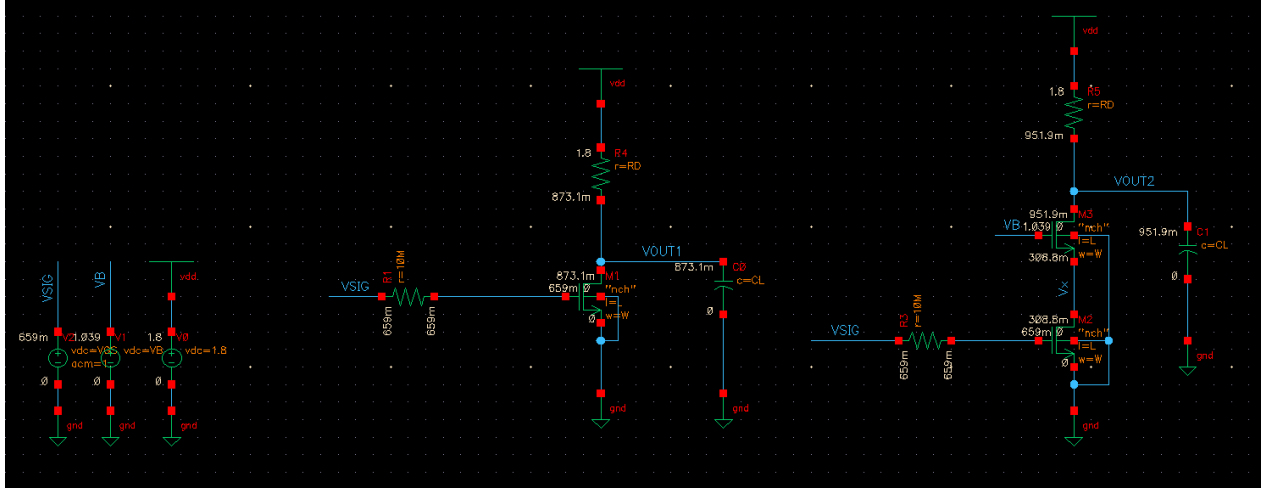


Figure 14: CS amplifier (M1, left) and cascode amplifier (M2, M3, right) with resistive loads R_D , $C_L = 1$ fF, with annotated voltages.

3.2 DC Operating Point Results

The DC OP simulation was run with $V_{GS} = 659$ mV, $V_B = 1.039$ V, $R_D = 13.24$ k Ω , $W = 4.86$ μ m, $L = 0.35$ μ m for all three transistors.

Global Variables	
☒	CL
☒	L
☒	RD
☒	VB
☒	W
☒	VGS
Click to add variable	

Figure 15: Variables used

ITLAB03_3:1	OP("M1" "id")	68.27u	ITLAB03_3:1	OP("M2" "id")	62.3u	ITLAB03_3:1	OP("M3" "id")	62.38u
ITLAB03_3:1	OP("M1" "vgs")	659m	ITLAB03_3:1	OP("M2" "vgs")	659m	ITLAB03_3:1	OP("M3" "vgs")	730.3m
ITLAB03_3:1	OP("M1" "vds")	896.2m	ITLAB03_3:1	OP("M2" "vds")	310.7m	ITLAB03_3:1	OP("M3" "vds")	664.4m
ITLAB03_3:1	OP("M1" "vth")	460m	ITLAB03_3:1	OP("M2" "vth")	464.6m	ITLAB03_3:1	OP("M3" "vth")	548.2m
ITLAB03_3:1	OP("M1" "vdsat")	152.8m	ITLAB03_3:1	OP("M2" "vdsat")	149.8m	ITLAB03_3:1	OP("M3" "vdsat")	151.6m
ITLAB03_3:1	OP("M1" "gm")	682u	ITLAB03_3:1	OP("M2" "gm")	637.5u	ITLAB03_3:1	OP("M3" "gm")	657.7u
ITLAB03_3:1	OP("M1" "gds")	8.61u	ITLAB03_3:1	OP("M2" "gds")	15.93u	ITLAB03_3:1	OP("M3" "gds")	9.066u
ITLAB03_3:1	OP("M1" "gmb")	178.4u	ITLAB03_3:1	OP("M2" "gmb")	167.4u	ITLAB03_3:1	OP("M3" "gmb")	155u
ITLAB03_3:1	OP("M1" "cgs")	-3.546f	ITLAB03_3:1	OP("M2" "cgs")	-4.046f	ITLAB03_3:1	OP("M3" "cgs")	-3.502f
ITLAB03_3:1	OP("M1" "cgd")	-2.321f	ITLAB03_3:1	OP("M2" "cgd")	-2.355f	ITLAB03_3:1	OP("M3" "cgd")	-2.321f
ITLAB03_3:1	OP("M1" "cgs")	-12.68f	ITLAB03_3:1	OP("M2" "cgs")	-12.69f	ITLAB03_3:1	OP("M3" "cgs")	-12.59f
ITLAB03_3:1	OP("M1" "csb")	-7.012f	ITLAB03_3:1	OP("M2" "csb")	-7f	ITLAB03_3:1	OP("M3" "csb")	-6.239f
ITLAB03_3:1	OP("M1" "region")	2	ITLAB03_3:1	OP("M2" "region")	2	ITLAB03_3:1	OP("M3" "region")	2

(a) M1 Parameters

(b) M2 Parameters

(c) M3 Parameters

Figure 16: Operating point parameters for transistors M1, M2, and M3.

Table 11: DC operating-point parameters for M1 (CS), M2 (cascode input), M3 (cascode device) – resistive-load design.

Parameter	M1	M2	M3
I_D (μA)	68.27	62.30	62.38
V_{GS} (mV)	659.0	659.0	730.3
V_{DS} (mV)	896.2	310.7	664.4
V_{TH} (mV)	460.0	464.6	548.2
V_{DSAT} (mV)	152.8	149.8	151.6
g_m ($\mu\text{A}/\text{V}$)	682.0	637.5	657.7
g_{ds} ($\mu\text{A}/\text{V}$)	8.610	15.93	9.066
g_{mb} ($\mu\text{A}/\text{V}$)	178.4	167.4	155.0
C_{db} (fF)	3.546	4.046	3.502
C_{gd} (fF)	2.321	2.355	2.321
C_{gs} (fF)	12.68	12.69	12.59
C_{sb} (fF)	7.012	7.000	6.239
Region	Sat (2)	Sat (2)	Sat (2)

Table 12: DC node voltages and bias (resistive-load design).

Node	Value
V_{OUT1} (M1 drain)	896.2 mV
V_{OUT2} (M3 drain)	975.1 mV
V_X (M2 drain / M3 source)	310.7 mV
V_B (M3 gate)	1.039 V
R_D	13.24 k Ω

3.2.1 Design Validation

Table 13: Effective overdrive at the operating point, compared to the 200 mV design target.

	M1	M2	M3
$V_{ov} = V_{GS} - V_{TH}$ (mV)	199.0	194.4	182.1

3.3 Saturation Check

Table 14: Saturation margin ($V_{DS} - V_{DSAT}$) – resistive-load design.

	M1	M2	M3
V_{DS} (mV)	896.2	310.7	664.4
V_{DSAT} (mV)	152.8	149.8	151.6
Margin (mV)	743.4	160.9	512.8

Comments

Q: Check that all transistors operate in saturation.

Confirmed: all three are **region=2**, and $V_{DS} > V_{DSAT}$ with healthy positive margins for each device.

3.4 AC Analysis

3.4.1 Hand Analysis

CS Amplifier (M1).

$$r_{o1} = \frac{1}{g_{ds1}} = 116.14 \text{ k}\Omega, \quad R_{out1} = r_{o1} \parallel R_D = 11.89 \text{ k}\Omega \quad (15)$$

$$A_{v1} = -g_{m1} R_{out1} = -682.0 \mu\text{A/V} \times 11.89 \text{ k}\Omega = -8.108 \quad (18.18 \text{ dB}) \quad (16)$$

$$C_{in,1} = C_{gs1} + C_{gd1}(1 + |A_{v1}|) = 12.68 \text{ fF} + 2.321 \text{ fF}(9.108) = 33.82 \text{ fF} \quad (17)$$

$$BW_1 = \frac{1}{2\pi R_s C_{in,1}} = \frac{1}{2\pi(10 \text{ M}\Omega)(33.82 \text{ fF})} = 470.7 \text{ kHz} \quad (18)$$

(The output-node pole, $1/(2\pi R_{out1}(C_L + C_{db1} + C_{gd1})) \approx 1.95 \text{ GHz}$, remains far higher and does not limit the bandwidth.)

$$GBW_1 = |A_{v1}| \cdot BW_1 = 3.817 \text{ MHz}, \quad UGF_1 \approx BW_1 \sqrt{A_{v1}^2 - 1} = 3.787 \text{ MHz} \quad (19)$$

Cascode Amplifier (M2, M3).

$$r_{o2} = 62.78 \text{ k}\Omega, \quad r_{o3} = 110.31 \text{ k}\Omega, \quad g_{m3} + g_{mb3} = 812.7 \mu\text{A/V} \quad (20)$$

$$R_{in,source} = \frac{R_D + r_{o3}}{1 + (g_{m3} + g_{mb3})r_{o3}} = \frac{13.24 \text{ k} + 110.31 \text{ k}}{1 + 89.65} = 1.363 \text{ k}\Omega \quad (21)$$

$$R_{VX} = R_{in,source} \parallel r_{o2} = 1.334 \text{ k}\Omega \quad (22)$$

$$A_{v,local} = -g_{m2} R_{VX} = -637.5 \mu\text{A/V} \times 1.334 \text{ k}\Omega = -0.850 \quad (23)$$

$$C_{in,2} = C_{gs2} + C_{gd2}(1 + 0.850) = 12.69 \text{ fF} + 2.355 \text{ fF}(1.850) = 17.05 \text{ fF} \quad (24)$$

$$BW_2 = \frac{1}{2\pi R_s C_{in,2}} = \frac{1}{2\pi(10 \text{ M}\Omega)(17.05 \text{ fF})} = 933.4 \text{ kHz} \quad (25)$$

$$R_{out,intrinsic} = r_{o2} + r_{o3} + (g_{m3} + g_{mb3})r_{o2}r_{o3} = 5.801 \text{ M}\Omega, \quad R_{out2} = R_{out,intrinsic} \parallel R_D = 13.21 \text{ k}\Omega \quad (26)$$

$$A_{v2} = -g_{m2} R_{out2} = -637.5 \mu\text{A/V} \times 13.21 \text{ k}\Omega = -8.421 \quad (18.51 \text{ dB}) \quad (27)$$

$$GBW_2 = |A_{v2}| \cdot BW_2 = 7.858 \text{ MHz}, \quad UGF_2 \approx BW_2 \sqrt{A_{v2}^2 - 1} = 7.806 \text{ MHz} \quad (28)$$

3.4.2 Simulation Results

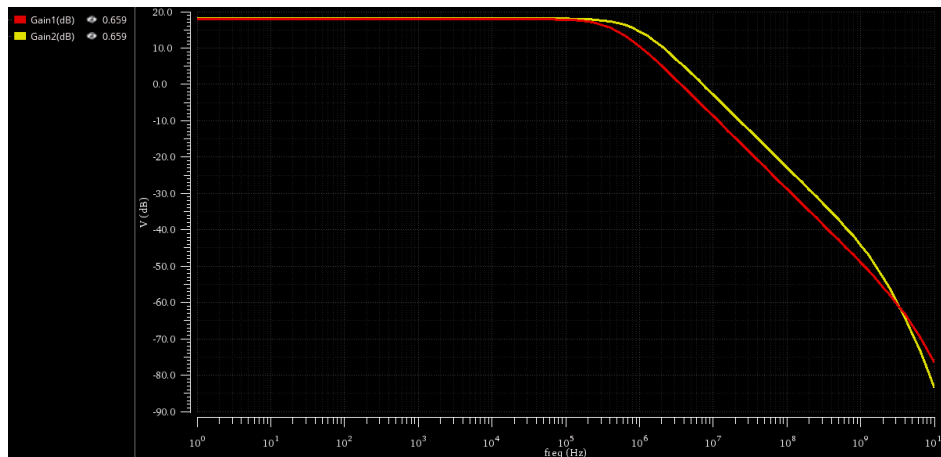


Figure 17: Magnitude in dB CS amplifier (Gain 1, red) and cascode amplifier (Gain 2, yellow).

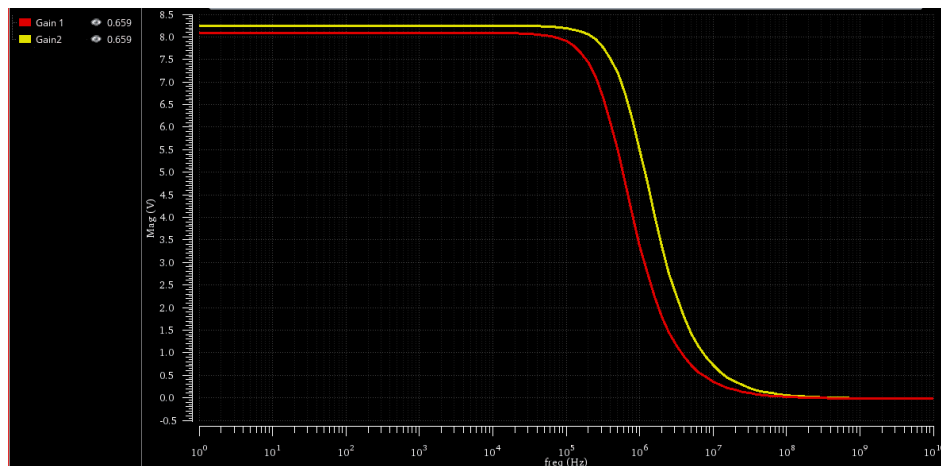


Figure 18: Linear Magnitude CS amplifier (Gain 1, red) and cascode amplifier (Gain 2, yellow).

3.4.3 Comparison: Simulation vs. Hand Analysis

Test	Output	Nominal
ITi:LAB03_3:1	Gain 1	✗
ITi:LAB03_3:1	Gain1(dB)	✗
ITi:LAB03_3:1	DC Gain1 (dB)	18.18
ITi:LAB03_3:1	DC Gain1	8.105
ITi:LAB03_3:1	BW1	460.5k
ITi:LAB03_3:1	GBW1	3.741M
ITi:LAB03_3:1	UGF_1	3.742M
ITi:LAB03_3:1	Gain2	✗
ITi:LAB03_3:1	Gain2(dB)	✗
ITi:LAB03_3:1	DC Gain2(db)	18.34
ITi:LAB03_3:1	DC Gain 2	8.262
ITi:LAB03_3:1	BW2	895k
ITi:LAB03_3:1	GBW2	7.412M
ITi:LAB03_3:1	UGF_2	7.416M

Figure 19: Simulation results

Table 15: CS amplifier (M1), resistive load: simulation vs. hand analysis.

Quantity	Hand Analysis	Simulation	Error (%)
DC Gain (V/V)	8.108	8.105	0.04
DC Gain (dB)	18.18	18.18	0.00
BW	470.7 kHz	460.5 kHz	2.22
GBW	3.817 MHz	3.741 MHz	2.03
UGF	3.787 MHz	3.742 MHz	1.20

Table 16: Cascode amplifier (M2, M3), resistive load: simulation vs. hand analysis.

Quantity	Hand Analysis	Simulation	Error (%)
DC Gain (V/V)	8.421	8.262	1.93
DC Gain (dB)	18.51	18.34	0.93
BW	933.4 kHz	895.0 kHz	4.29
GBW	7.858 MHz	7.412 MHz	6.02
UGF	7.806 MHz	7.416 MHz	5.26

Comments

Q: Comment on the results.

The simulation data clearly demonstrates the secondary major advantage of the cascode topology: **bandwidth extension via Miller-effect suppression**.

When driven by a high-impedance source ($R_{sig} = 10\text{ M}\Omega$), the common-source amplifier suffers a severe bandwidth penalty because its C_{gd1} is multiplied by the full voltage gain of the stage ($1 + |A_{v1}| \approx 9.1\times$), creating a large effective input capacitance ($\approx 33.8\text{ fF}$) that sets a comparatively low-frequency dominant pole (460.5 kHz).

In contrast, the cascode amplifier delivers essentially the same total DC voltage gain (18.34 dB vs. 18.18 dB) but splits the voltage swing across two devices: M2's drain is loaded by the low impedance looking into M3's source ($R_{VX} \approx 1.33\text{ k}\Omega$), restricting M2's local voltage gain to only ≈ 0.85 . This prevents any significant Miller multiplication of C_{gd2} , so the total input capacitance at M2's gate is roughly halved ($\approx 17.1\text{ fF}$). The result is that the cascode's bandwidth (895.0 kHz) is $\approx 1.94\times$ that of the CS stage, and its GBW (7.412 MHz) is $\approx 1.98\times$ higher – essentially the cascode "buys" close to double the speed at matched gain, the opposite trade-off from Part 2 where gain was traded for bandwidth at fixed GBW. The hand-analysis errors are somewhat larger here (up to 6 % on GBW) than in the CS case, since the cascode's bandwidth model compounds several approximated quantities ($R_{in,source}$, R_{VX} , the local Miller gain, and the neglected second pole at V_X), each contributing a small additional deviation.